

1 Exact determinant of the polynomial–exponential period comparison

Fix the original monic cyclic polynomial

$$\chi(S) = S^q + \sum_{a=0}^{q-1} c_a S^a, \quad q \geq 1, \quad d = q + 1, \quad \Phi_t(S) = \frac{S^d}{d} + \sum_{a=0}^{q-1} \frac{c_a S^{a+1}}{a+1} - tS.$$

Fix $u \in \mathbb{C}^*$ and one argument of u . All the following comparisons retain this u , the original coordinate S , and the ordered remainder basis $1, S, \dots, S^{q-1}$. Set

$$D_{u,t}P = uP' + (\chi - t)P, \quad H_{u,t} = \mathbb{C}[S]/D_{u,t}\mathbb{C}[S].$$

This is a quotient of coefficient vector spaces. Let $\text{red}_{u,t}$ be its unique representative of degree less than q . Let $A(t)$ be the multiplication matrix of S on $\mathbb{C}[S]/(\chi - t)$ in the same ordered basis, so

$$A(t)e_b = e_{b+1} \quad (0 \leq b < q-1), \quad A(t)e_{q-1} = (-c_0 + t)e_0 - \sum_{a=1}^{q-1} c_a e_a. \quad (\text{XD1})$$

The empty sum convention in this display retains the case $q = 1$.

Use precisely the oriented rays and differences

$$\theta_j = \frac{\arg u + \pi}{d} + \frac{2\pi j}{d}, \quad \ell_j : S = re^{i\theta_j}, \quad 0 \leq r < \infty, \quad \Gamma_j = -\ell_0 + \ell_j \quad (1 \leq j \leq q). \quad (\text{XD2})$$

Write

$$\Pi_{j,b}(u, c, t) = \int_{\Gamma_j} e^{\Phi_t(S)/u} S^b dS, \quad 1 \leq j \leq q, \quad 0 \leq b < q. \quad (\text{XD3})$$

1.1 Uniform estimates on the fixed unbounded contours

For a compact set $K \subset \mathbb{C}^{q+1}$ of the parameters $(c_0, \dots, c_{q-1}, t)$, put

$$M_K = \sum_{a=0}^{q-1} \frac{\sup_K |c_a|}{a+1} + \sup_K |t|, \quad R_K = \max\{1, 2dM_K\}, \quad \alpha = \frac{1}{2d|u|}. \quad (\text{XD4})$$

For every ray in (XD2) its leading term satisfies $S^d/(du) = -r^d/(d|u|)$. For $r \geq 1$ every other exponent has degree at most $d-1$, so

$$\text{Re} \frac{\Phi_t(re^{i\theta_j})}{u} \leq -\frac{r^d}{d|u|} + \frac{M_K r^{d-1}}{|u|} \leq -\alpha r^d \quad (r \geq R_K). \quad (\text{XD5})$$

Every parameter derivative of the integrand is its original exponential times a polynomial in S : a derivative in c_a contributes $S^{a+1}/((a+1)u)$ and a derivative in t contributes $-S/u$. For any fixed derivative order its absolute value on the tail is bounded by $Cr^L e^{-\alpha r^d}$, with C, L uniform on K . The required majorant is integrable, since substitution $x = \alpha r^d$ gives

$$\int_0^\infty r^L e^{-\alpha r^d} dr = \frac{1}{d} \alpha^{-(L+1)/d} \Gamma\left(\frac{L+1}{d}\right) < \infty. \quad (\text{XD6})$$

The remaining finite interval is uniformly bounded by continuity. Thus (XD3) converges absolutely and is entire in all the lower coefficients and t ; every parameter derivative may be taken under the integral. The same bound gives $e^{\Phi_t(S)/u} P(S) \rightarrow 0$ at the infinite endpoint of each ray, for every polynomial P . Consequently

$$\int_{\Gamma_j} e^{\Phi_t(S)/u} D_{u,t}P dS = u \int_{\Gamma_j} d(e^{\Phi_t(S)/u} P) = 0. \quad (\text{XD7})$$

Indeed the integral of that derivative on each outward ray is $-e^{\Phi_t(0)/u} P(0) = -P(0)$; the coefficients $-1, +1$ of the two rays cancel these values. These estimates handle the noncompact contours directly, including parameter values at which critical points coincide.

1.2 An exact trace comparison for twisted reduction

For $1 \leq m \leq q$ and $0 \leq b < q$, ordinary monic division gives unique polynomials $Q_{m,b}, R_{m,b}$ satisfying

$$S^{m+b} = (\chi - t)Q_{m,b} + R_{m,b}, \quad \deg R_{m,b} < q. \quad (\text{XD8})$$

If $m + b < q$, take $Q_{m,b} = 0$. Otherwise $\deg Q_{m,b} = m + b - q \leq q - 1$, and the differential itself gives

$$\text{red}_{u,t}(S^{m+b}) = R_{m,b} - uQ'_{m,b}. \quad (\text{XD9})$$

Both terms on the right have degree below q . In every nonzero case,

$$\deg Q'_{m,b} \leq m + b - q - 1 \leq b - 1. \quad (\text{XD10})$$

Thus the coefficient of S^b in the derivative correction vanishes. Let $T_m(u, c, t)$ have column b equal to the coordinates of the left side of (XD9). The columns $R_{m,b}$ are exactly those of $A(t)^m$. Although (XD9) retains a possibly nonzero correction matrix, (XD10) proves the exact trace identity

$$\text{Tr } T_m(u, c, t) = \text{Tr } A(t)^m \quad (1 \leq m \leq q). \quad (\text{XD11})$$

Set $B_a = T_{a+1}/((a+1)u)$ for $0 \leq a < q$. Differentiation of (XD3), followed by (XD7)–(XD9), now proves

$$\partial_{c_a} \Pi = \Pi B_a, \quad \partial_t \Pi = -\Pi A(t)/u, \quad \text{Tr } B_a = \frac{\text{Tr } A(t)^{a+1}}{(a+1)u}. \quad (\text{XD12})$$

All the coefficient matrices here have polynomial entries in c, t for this fixed nonzero u .

1.3 The determinant, including every monomial factor

Define the scalar polynomial in the original coefficients

$$\mathcal{F}(c, t) = \text{Tr } \Phi_t(A(t)) = \frac{\text{Tr } A(t)^{q+1}}{q+1} + \sum_{a=0}^{q-1} \frac{c_a \text{Tr } A(t)^{a+1}}{a+1} - t \text{Tr } A(t). \quad (\text{XD13})$$

There is no choice or enumeration of roots in this definition. For any differentiable matrix A and integer $n \geq 1$, the product rule and cyclic invariance of trace give

$$d \text{Tr } A^n = \sum_{r=0}^{n-1} \text{Tr}(A^r(dA)A^{n-1-r}) = n \text{Tr}(A^{n-1}dA). \quad (\text{XD14})$$

Because $A(t)$ acts on the actual quotient by $\chi - t$, it satisfies $\chi(A(t)) - tI_q = 0$. Applying (XD14) to (XD13) therefore gives

$$d\mathcal{F} = \sum_{a=0}^{q-1} \frac{\text{Tr } A(t)^{a+1}}{a+1} dc_a - \text{Tr } A(t) dt + \text{Tr}((\chi(A(t)) - tI_q)dA(t)) = \sum_{a=0}^{q-1} \frac{\text{Tr } A(t)^{a+1}}{a+1} dc_a - \text{Tr } A(t) dt. \quad (\text{XD15})$$

At $c = t = 0$ the period matrix is

$$\Pi_{j,l-1}^{\text{mon}}(u) = a_l(u) e^{il\theta_0} (\zeta^{jl} - 1), \quad a_l(u) = \frac{(d|u|)^{l/d}}{d} \Gamma(l/d), \quad \zeta = e^{2\pi i/d}, \quad 1 \leq j, l \leq q. \quad (\text{XD16})$$

This follows by substituting $x = r^d/(d|u|)$ on each oriented ray in (XD3). In particular all $a_l(u)$ are positive real numbers. For $V_{jl} = \zeta^{jl} - 1$ one may include $j = 0$ in the Gram sum because its summand is zero. The complete root-of-unity sums give

$$(V^*V)_{lm} = \sum_{j=0}^{d-1} (\zeta^{-jl} - 1)(\zeta^{jm} - 1) = d\delta_{lm} + d. \quad (\text{XD17})$$

Here $1 \leq l, m < d$, so the two individual nonconstant sums vanish, and the first sum is $d\delta_{lm}$. The all-ones rank-one matrix has eigenvalue q on the all-ones vector and zero on its $(q-1)$ -dimensional perpendicular complement. It follows that

$$\det(V^*V) = d^q(1+q) = d^d > 0, \quad C_q(u) := |\det \Pi^{\text{mon}}(u)|^2 = d^d \prod_{l=1}^q \left[\frac{(d|u|)^{l/d}}{d} \Gamma(l/d) \right]^2. \quad (\text{XD18})$$

The complex determinant itself retains the oriented phase: $\det \Pi^{\text{mon}} = \det V \prod_{l=1}^q a_l(u) e^{i\theta_0}$.

We claim the exact global formula

$$\boxed{\det \Pi(u, c, t) = \det \Pi^{\text{mon}}(u) \exp(\mathcal{F}(c, t)/u)}. \quad (\text{XD19})$$

To prove it without assuming invertibility along the path, join $(0, 0)$ to (c, t) by (sc, st) , $0 \leq s \leq 1$. Equations (XD12) give $\Pi' = \Pi B(s)$, with continuous $B(s)$, and (XD15) gives $\text{Tr } B(s) = \mathcal{F}'(sc, st)/u$. For every square matrix, including a singular one,

$$\frac{d}{ds} \det \Pi = \text{Tr}(\text{adj}(\Pi) \Pi B) = \det \Pi \text{Tr } B. \quad (\text{XD20})$$

Consequently $\det \Pi \exp(-\mathcal{F}/u)$ has derivative zero. At the starting point A is nilpotent and $\mathcal{F}(0, 0) = 0$, so its value is $\det \Pi^{\text{mon}}(u)$. This proves (XD19). Equation (XD18) proves its nonvanishing for every c, t , including every repeated-root fibre. In particular the period map

$$H_{u,t} \longrightarrow \mathbb{C}^q, \quad [P] \longmapsto \Pi \text{red}_{u,t}(P), \quad (\text{XD21})$$

is an isomorphism with inverse given by Π^{-1} followed by the specified degree-below- q section.

1.4 Exact determinant transport to the arithmetic representative

Retain the original positive Gram G_N on the cyclic coefficient space E , its determinant $V_N = \det G_N$, and the original maps $r_N : E \longrightarrow \mathcal{B}^{\otimes k}$ and j_E on the admitted test-function domain with $j_E r_N = I_E$. Identify E with $H_{u,t}$ by their specified ordered remainder bases. Then the actual maps

$$r_N(E) \longrightarrow \mathbb{C}^q, \quad x \longmapsto \Pi j_E x, \quad \mathbb{C}^q \longrightarrow r_N(E), \quad y \longmapsto r_N \Pi^{-1} y \quad (\text{XD22})$$

are inverse: their composites are respectively $r_N j_E$ on $r_N(E)$ and $\Pi(j_E r_N) \Pi^{-1}$ on $\mathbb{C}^q$. On the full admitted domain the first observation has precisely $\ker j_E$ as its kernel; all original theta boundaries killed by j_E remain in that kernel.

With the explicitly fixed coordinate metric I_q on the target, put

$$H_\Pi = \Pi^* \Pi, \quad B_N^{\text{per}} = G_N^{-1} H_\Pi. \quad (\text{XD23})$$

These operators satisfy $G_N B_N^{\text{per}} = (B_N^{\text{per}})^* G_N = H_\Pi > 0$. Taking determinants in (XD19) gives the exact identities

$$\det H_\Pi = C_q(u) \exp\left(2 \text{Re} \frac{\mathcal{F}(c, t)}{u}\right), \quad (\text{XD24})$$

$$\boxed{\log \det B_N^{\text{per}} = \log C_q(u) + 2 \text{Re} \frac{\mathcal{F}(c, t)}{u} - \log V_N}. \quad (\text{XD25})$$

Every determinant in (XD24)–(XD25) is positive real; these logarithms therefore require no branch choice. At the same u, c, t for two admitted degrees $i \leq j$, subtraction proves

$$\boxed{\log \det B_j^{\text{per}} - \log \det B_i^{\text{per}} = \log(V_i/V_j)}. \quad (\text{XD26})$$

Thus the common polynomial–exponential factor is now calculated exactly while the full original endpoint cost is retained by the displayed pair of positive comparison operators. No uniform upper estimate on V_i/V_j is inferred from this determinant identity.